Q1. DIRECTIONS *for question 1:* Type in your answer in the input box provided below the question.

In how many ways can an amount of ₹100 be paid using exactly 27 coins of denominations ₹1, ₹2 and ₹5, such that at least one coin of each denomination is used?

Q2. DIRECTIONS *for question 2:* Select the correct alternative from the given choices.

The number of real roots of the equation $\frac{A}{x-1}+\frac{B}{x+2}=1$, where *A* and *B* are both negative real numbers, is

a) 0.

b) 1.

c) 2.

d) Cannot be determined

Q3. DIRECTIONS *for question 3:* Type in your answer in the input box provided below the question.
What are the last two digits of $849^{23} + 521^{63}$?

Q4. DIRECTIONS *for questions 4 to 10:* Select the correct alternative from the given choices.

The total cost of 2 pencils, 5 erasers and 7 sharpeners is ₹30, while 3 pencils and 5 sharpeners cost ₹15 more than 6 erasers. By what amount (in ₹) does the cost of 39 erasers and 1 sharpener exceed the cost of 6 pencils?

- a) 20
- b) 30
- c) 27
- d) Cannot be determined

Q5.

DIRECTIONS *for questions 4 to 10:* Select the correct alternative from the given choices.

What is the number of integral values of x which satisfy both the inequalities $8x^2 + 6x - 27 < 0$ and $-x^2 + 11x + 80 > 0$?

- a) 1
- b) 3
- c) 2
- d) 4

Q6.

DIRECTIONS *for questions 4 to 10:* Select the correct alternative from the given choices.

If one of the sides of a right-angled triangle with integer sides is 15 cm, find the maximum possible area of the triangle (in sq.cm).

- a) 54
- b) 187.5
- c) 720
- d) 840

Q7.

DIRECTIONS *for questions 4 to 10:* Select the correct alternative from the given choices.

A certain number of bacterial cells are placed in a petri dish and every hour exactly *k*% of the bacteria that are present at the beginning of the hour perish. If it was noted that the number of bacteria that perished in the first two hours is the same as the number of all the bacteria that perished after the first two hours, approximately what percentage of the initial bacteria were alive after the third hour?

a) 26.24%

b) 29.37%

c) 42.10%

d) 35.36%

Q8.

DIRECTIONS *for questions 4 to 10:* Select the correct alternative from the given choices.

If P(a^2, 2a) is a point on the line segment joining the points A(2, 0) and B(0, 4), what is the ratio of the distances AP and PB ?

a)
$\sqrt{2} : 1$

b) 1 : 1

c) 2 : 1

d)
$1 : \sqrt{3}$

Q9.

DIRECTIONS *for questions 4 to 10:* Select the correct alternative from the given choices.

Yesterday, A and B sold a distinct number of mangoes each and both of them received the same amount of money. Today, A increased the price of his mangoes by a certain percentage, while B decreased the price of his mangoes by the same percentage. If each of them sold the same number of mangoes today as they did the previous day and the sum received by A is twice the sum received by B, find the percentage by which A increased the price of his mangoes.

a) $33\frac{1}{3}\%$

b) 25%

c) 50%

d) Cannot be determined

Q10.

DIRECTIONS *for questions 4 to 10:* Select the correct alternative from the given choices.

Tap A and Tap B, when opened simultaneously, completely fill an empty tank in 20 minutes. If Tap A alone can fill three-quarters of the tank in 33 minutes, how long will it take Tap B alone to completely fill the tank?

- a) 35 minutes
- b) 34.33 minutes
- c) 36.67 minutes
- d) 34 minutes

Q11.

DIRECTIONS *for question 11:* Type in your answer in the input box provided below the question.

Let $f_{n+1}(x) = f_n(x) + 1$, if n is a multiple of 3.

$= f_n(x) – 1$, otherwise.

If $f_1(1) = 0$, then find the value of $f_{50}(1)$.

Q13. DIRECTIONS *for question 13:* Type in your answer in the input box provided below the question.

There are 32 teams participating in a tournament, where each team plays exactly one match against every other team. For a win, a team is awarded two points; for a draw, each of the two teams gets one point; for a loss, no points are awarded. If the team that scored more number of points in the tournament than any other team was declared the winner of the tournament, the winner must have scored at least

Q14. DIRECTIONS *for questions 14 and 15:* Select the correct alternative from the given choices.
Given that $-8 \le x \le -5$, $-4 \le y < 2$ and $-1 \le z \le 5$, which of the following statements is/are always true?

a) $9z + 5 > x + 2y$

b) $6z > 5x + 9y$

c) $2(x + y) < 5z$

d) More than one of the above

Q15. DIRECTIONS *for questions 14 and 15:* Select the correct alternative from the given choices.

What is the area of the circumcircle of a right-angled triangle whose base is 5 cm and height, 12 cm?

a)
42.25π sq.cm.

b)
144π sq.cm.

c)
169π sq.cm.

d)
84.5π sq.cm.

Q16. DIRECTIONS *for question 16:* Type in your answer in the input box provided below the question.

If k is a natural number such that $f(f(f(f(k)))) = 1$ and $k > f(k) > f(f(k)) > f(f(f(k))) > 1$, what is the least number of digits that k can have?

Q17. DIRECTIONS *for question 17:* Select the correct alternative from the given choices.

For a natural number m, which of the following could be the value of $f(f(m - f(m)))$?

- a) 12
- b) 15
- c) 18
- d) 21

Q18. DIRECTIONS *for question 18:* Select the correct alternative from the given choices.

If the product of a surd $a + \sqrt{b}$ and its conjugate is 33 and the product *ab* is 18, which of the following can be the value of *a*?

- a) 2
- b) 3
- c) 6
- d) 9

Q19. DIRECTIONS *for question 19:* Type in your answer in the input box provided below the question.

Find the number of distinct terms in the expansion of $(x + y + z + w)^{10}$.

Q20. DIRECTIONS *for question 20:* Select the correct alternative from the given choices.

If three runners, A, B and C, start simultaneously from the same point and run around a circular track of length 500 m, in the same direction, at speeds of 5 kmph, 8 kmph and 15 kmph respectively, what is the time taken by them to meet for the first time?

- a) 60 minutes
- b) 30 minutes
- c) 15 minutes
- d) 10 minutes

Q21. DIRECTIONS *for question 21:* Type in your answer in the input box provided below the question.

What is the sum of the digits of the smallest number which when divided by 7 leaves a remainder of 6, when divided by 8 leaves a remainder of 7 and when divided by 9 leaves a remainder of 1?

Q22. DIRECTIONS *for question 22:* Select the correct alternative from the given choices.

If the roots of the equation $(x - p)(x - q) + r = 0$ are 4 and 5, where p, q and r are non-zero positive integers, then what is the highest possible value of r?